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Integrating creativity and artificial intelligence capability in entrepreneurial ventures

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Abstract

Purpose – While the literature on artificial intelligence (AI) capability is expanding, gaps remain in understanding how this capability is internally developed in technology-based startups (TBS) across different life cycle phases. This study, grounded in the resource orchestration theory (ROT), investigates the pathway through which TBS use organizational creativity to build AI capability and achieve performance.

Design/methodology/approach – A conceptual framework based on ROT emphasizes the role of organizational creativity in the structuring and bundling processes. Data were collected through a survey of 166 managers and employees of TBS operating in Brazil and international markets, using multiple linear regressions and the Sobel test for analysis. The study validated the AI capability scale in the TBS context.

Findings – AI capability fully mediates the relationship between organizational creativity and performance, confirming that organizational creativity is a critical resource for AI capability development. These findings advance ROT by deepening the understanding of how AI capability is developed in TBS. The study offers a dynamic, process-based view of performance trajectories in TBS, demonstrating that the synchrony between creativity and AI capability creates a cyclical process, maximizing company performance.

Originality/value – This research identifies an alternative pathway for TBS to develop AI capability and achieve performance, highlighting the synchronization and co-evolution of resources and capabilities. It provides novel insights into AI capability's mediating role and expands understanding of resource management in TBS across life cycle phases.

Keywords Organizational creativity, Artificial intelligence capability, Organizational performance, Life cycle, Resource orchestration

Paper type Research paper

1. Introduction

Artificial Intelligence (AI) Capability is the ability of a firm to select, orchestrate, and leverage its AI-specific resources (Mikalef and Gupta, 2021). The literature (Mikalef and Gupta, 2021; Li *et al.*, 2022; Ameen *et al.*, 2024) states the path of AI Capability to organizational creativity to performance. It assumes that AI Capability is already implemented beyond an experimental stage in established companies, leaving gaps in understanding the internal processes required for its creation (Sjödin *et al.*, 2021; Chowdhury *et al.*, 2023; Madanaguli *et al.*, 2024). Think of a startup without AI Capability or in an incumbent company planning to develop AI Capability. If organizational creativity is a strategic resource that is a precursor to capability development (Javidan, 1998; de Vasconcellos *et al.*, 2021), then in the early stages of the

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organizational life cycle or of the capability life cycle, creativity is an antecedent of AI Capability.

In this context, we argue that organizational creativity is instrumental in developing AI Capability. Creativity aids in acquiring, developing, and integrating AI resources, which are essential for forming capabilities (Grilli and Pedota, 2024). Using Resource Orchestration Theory (ROT) (Sirmon *et al.*, 2007, 2011), we provide a framework for understanding how creativity forms capabilities like AI Capability and performance. Importantly, ROT underscores the significance of resource management processes—structuring, bundling, and leveraging—in converting resources into capabilities that generate performance (Sirmon *et al.*, 2007, 2011; Ireland *et al.*, 2003). In this way, we focus on technology-based startups (TBS) at different life cycle stages. Also, we focus on Brazilian startups recognized for addressing local challenges and increasingly competing in global markets (Cahen and Borini, 2020) leading AI adoption in Latin America (Sebrae Startups, 2023). Given the limited resources typical of TBS startups and Brazilian context, organizational creativity plays a pivotal role in developing capabilities like AI Capability, even during the early phases of their life cycle.

In this way, the article addresses the following research question: What is the association between organizational creativity and AI Capability, and performance in TBS? The objective is to demonstrate the pathway through which TBS uses organizational creativity to build AI Capability and achieve performance across various life cycle phases. By focusing on TBS in Brazil, this study addresses a critical gap in the literature. Previous research has predominantly examined AI Capability in established firms in the United States and China (Mikalef and Gupta, 2021; Li *et al.*, 2022; Ameen *et al.*, 2024), asking for further studies in other organizational life cycles, with constrained resources (like TBS) (Ameen *et al.*, 2024) and in underrepresented contexts.

The article defending creativity as an antecedent of AI Capability offers literature implications and insights for managers and policymakers. Using ROT extends the understanding of AI Capability (Mikalef and Gupta, 2021), showing how it changes roles addressing the underexplored dimension of organizational cycle stages, especially in TBS companies (Badrinarayanan *et al.*, 2019; Symeonidou and Nicolaou, 2018). To the management practice, it identifies an alternative pathway for TBS to develop AI Capability and achieve performance, offering insights into resource synchronization and the co-evolution of AI Capability with organizational creativity, as predicted by ROT (Sirmon *et al.*, 2007, 2011). For policymakers, we bring insights to implement policies that foster innovation and the development of AI technologies. Finally, we discuss the societal implications of the articulation of AI and creativity in resource-limited contexts.

2. Conceptual framework

2.1 Organizational performance from the resource orchestration theory

Resource Orchestration Theory (ROT) emphasizes the importance of managerial actions in transforming resources into capabilities to achieve organizational performance. As an extension of the Resource-Based View, ROT highlights that beyond the value, rarity, inimitability, and organization of resources, managerial action plays a critical role in resource management, capability creation, and performance enhancement (Badrinarayanan *et al.*, 2019; Hughes *et al.*, 2018; Sirmon *et al.*, 2007, 2011). ROT integrates resource management and asset orchestration dimensions. Resource management involves three key processes: *structuring*, *bundling*, and *leveraging* (Sirmon *et al.*, 2007, 2011). This study focuses on the structuring and bundling processes, highlighting organizational creativity as a crucial antecedent for AI Capability development. It also examines asset orchestration across TBS life cycle phases to explore how the initial, transition, and scalability stages influence AI Capability and performance, considering the role of creativity.

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Structuring refers to managing the resource portfolio, which evolves throughout a company's life cycle (Sirmon *et al.*, 2007, 2011; Carnes *et al.*, 2017). This process is particularly relevant for technology-based startups (TBS) during the initial, transition, and scalability phases (Picken, 2017a, b, 2017b). The *structuring* process encompasses all the resources available to TBS, including organizational creativity, tangible, intangible and human, and AI-related resources. Organizational creativity is pivotal for acquiring and internally developing resources (Sirmon *et al.*, 2007, 2011). For example, it helps managers acquire data, AI technologies, and basic resources (Glaveanu and Taillard, 2018) while enabling employees to recombine or develop these resources internally (Grilli and Pedota, 2024; Zahra, 2021). Organizational creativity integrates AI-related resources, shaping generative outputs and enabling effective AI training (Grilli and Pedota, 2024).

Once resources are developed through the structuring process, they enter the *bundling* process, where they are integrated to create specific capabilities (Chadwick *et al.*, 2015; Ireland *et al.*, 2003; Sirmon *et al.*, 2007). Bundling involves the subprocess of *pioneering* that is mainly influenced by creativity, as it drives the creation of new capabilities (Sirmon *et al.*, 2007) by integrating resources, enabling organizations to adapt to environmental changes (Badrinarayanan *et al.*, 2019) and enhance performance (Carnes *et al.*, 2017; Hughes *et al.*, 2018; Sirmon *et al.*, 2007). AI Capability, as a novel capability, is created through this subprocess. Here, creativity integrates various resources (Sirmon *et al.*, 2007)—tangible, intangible, and human AI resources (Mikalef and Gupta, 2021; Li *et al.*, 2022)—to build AI Capability.

Under the ROT framework, creativity contributes to both structuring resources and bundling them through pioneering. Once AI Capability is created, organizational creativity reenters the structuring process, ensuring continuous innovation. When a capability is implemented, it begins to intervene in the relationship between resources and performance (Sirmon *et al.*, 2007, 2011). Organizational performance and resource management are further influenced by variations in asset orchestration (Badrinarayanan *et al.*, 2019), including: *breadth*, addressing the entire organizational scope; *depth*, encompassing all managerial hierarchical levels; and *life cycle*, considering resource orchestration throughout the company's life cycle (Chadwick *et al.*, 2015; Sirmon *et al.*, 2011). The extent to which a TBS progresses through its life cycle to integrate creativity with AI resources and develop AI Capability warrants further examination (Badrinarayanan *et al.*, 2019; Symeonidou and Nicolaou, 2018).

2.2 Organizational creativity

Organizational creativity is defined as "the creation of a valuable and useful new product, service, idea, procedure, or process by individuals working together in a complex social system" (Woodman *et al.*, 1993, p. 293). Management scholars regard organizational creativity as an intangible, high-level resource that is valuable, rare, inimitable, and organized (de Vasconcellos *et al.*, 2019; 2024; Gao *et al.*, 2021; Ameen *et al.*, 2024), making it essential for both the early stages and the long-term survival of companies (Gao *et al.*, 2021; Zahra, 2021).

The broad conceptualization of organizational creativity allows it to be viewed as an accessible and actionable resource within organizations (de Vasconcellos *et al.*, 2019). This is particularly relevant for start-ups, where the creativity of entrepreneurs drives business development and helps overcome resource scarcity (Glaveanu and Taillard, 2018). In this context, organizational creativity facilitates access to new resources and their integration into the company's resource portfolio (Glaveanu and Taillard, 2018; Zahra, 2021). It also aids employees in developing or recombining resources internally (Zahra, 2021; Grilli and Pedota, 2024), including AI-related resources. Effectively combining organizational creativity with other resources like tangible, intangible, and human AI resources is essential for its successful application and enhancing performance (de Vasconcellos *et al.*, 2021; Grilli and Pedota, 2024).

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Furthermore, organizational creativity underpins capability development, fostering competencies and enhancing performance (Anderson *et al.*, 2014; de Vasconcellos *et al.*, 2021; 2024). The relationship between creativity and performance is a core topic in organizational creativity research and has been substantiated by several studies. For example, Mikalef and Gupta (2021), de Vasconcellos *et al.* (2021; 2024), and Weinzimmer *et al.* (2011) demonstrated a significant positive effect of organizational creativity on performance.

According to de Vasconcellos *et al.* (2021; 2024), companies that prioritize creativity report superior performance outcomes. Organizational creativity not only fosters competitive advantages (Weinzimmer *et al.*, 2011; Acar *et al.*, 2019) but also plays a subjective role in influencing entrepreneurial behavior and decision-making processes, ultimately influencing organizational performance (de Vasconcellos *et al.*, 2019). This evidence from the literature leads us to propose Hypothesis 1, which states:

H1. Organizational creativity is positively associated with organizational performance in TBS.

2.3 Artificial intelligence capability

Recent research has increasingly examined AI Capability (Mikalef and Gupta, 2021; Sjödin *et al.*, 2021; Li *et al.*, 2022; Madanaguli *et al.*, 2024; Ameen *et al.*, 2024). However, the understanding of how AI Capability is formed through the integration of resources is still developing (Sjödin *et al.*, 2021; Chowdhury *et al.*, 2023; Madanaguli *et al.*, 2024). Despite these efforts, the role of organizational creativity in building AI Capability remains underexplored (Acar *et al.*, 2019; Grilli and Pedota, 2024).

Javidan (1998) argues that capabilities are function-specific and are developed through the integration of resources. However, AI resources do not integrate autonomously (Mikalef and Gupta, 2021; Chowdhury *et al.*, 2023). AI encompasses diverse methodologies, tools, and applications (Abaddi, 2023) and achieve artificial autonomy (Abaddi, 2024) only after advancing beyond experimental stages (Mikalef and Gupta, 2021), requiring human intervention and creativity for their training and application (Zeng and Khan, 2019; de Vasconcellos *et al.*, 2021; Grilli and Pedota, 2024; Mariani and Dwivedi, 2024).

Thus, organizational creativity is essential for developing and integrating AI resources (Grilli and Pedota, 2024), as it underpins their adoption, implementation, and association with resource management (Sirmon *et al.*, 2007, 2011; Chowdhury *et al.*, 2023). The specific resources contributing to AI Capability include tangible, intangible, and human resources (Mikalef and Gupta, 2021). Tangible AI resources comprise data, technology, and basic resources such as time and financial resources, while intangible AI resources include inter-departmental coordination, organizational change capacity, and risk proclivity. Human AI resources encompass technical and business skills (see Mikalef and Gupta, 2021). Effectively orchestrating these resources collectively strengthens AI Capability (Mikalef and Gupta, 2021; Sirmon *et al.*, 2011).

Creativity plays a vital role in digital transformation, fostering the application of new technologies (Grilli and Pedota, 2024) and enabling the development of digital capabilities in organizations (de Vasconcellos *et al.*, 2021). Organizational creativity reflects entrepreneurial skills and collaboration (Acar *et al.*, 2019), which are crucial for generating capabilities to navigate complex environments (de Vasconcellos *et al.*, 2019). From the ROT perspective, organizational creativity is the resource that integrates tangible, intangible, and human-AI resources (Mikalef and Gupta, 2021; Grilli and Pedota, 2024), making it fundamental to building a new and novel capability (Sirmon *et al.*, 2007) like AI Capability.

While organizational creativity has been linked to fostering innovative, entrepreneurial, and digital capabilities (de Vasconcellos *et al.*, 2019; 2021), its role in developing AI Capability remains underexplored. This evidence from the literature leads us to propose Hypothesis 2, which states:

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H2. Organizational creativity is positively associated with AI Capability in TBS.

One of the most critical activities for entrepreneurs is the accumulation and combination of resources to achieve superior performance (Symeonidou and Nicolaou, 2018). Resource management, encompassing structuring, bundling, and their internal subprocesses, provides a framework for understanding how TBS utilize their resources to develop new capabilities and gain performance advantages (Sirmon *et al*., 2007, 2011). From the perspective of ROT, excellence in resource management processes can drive a company toward superior performance (Andersén, 2021). Notably, a company excelling in one of the three processes—structuring, bundling, or leveraging—can offset average performance in the others (Andersén, 2021; Sund *et al*., 2018). Thus, if a TBS excels in bundling through pioneering, its AI Capability is likely to contribute to superior performance.

ROT underscores the importance of evaluating the outcomes of resource bundling, specifically whether a capability translates into performance improvements (Sund *et al*., 2018). For instance, Chowdhury *et al*. (2023) demonstrated that AI Capability enhances organizational performance by optimizing operations and resources, automating tasks, reducing production time, improving delivery and responsiveness, and lowering operational costs. Similarly, Wamba-Taguimdje *et al*. (2020) highlighted that AI Capability positively impacts financial, marketing, administrative, and process performance. Additionally, Mikalef and Gupta (2021) found that AI Capability leads to superior performance. Despite these insights, the relationship between AI Capability and performance remains an ongoing area of research, particularly regarding AI's broader impact (Tran and Murphy, 2023) and AI Capability's role in TBS. Based on this, we propose the following hypothesis:

H3. AI Capability is positively associated with the organizational performance in TBS.

From the ROT perspective (Sirmon *et al*., 2007, 2011), capabilities like AI Capability, when developed and implemented, intervene in the relationship between resources and performance. The coexistence of creativity and AI Capability within structuring and bundling processes enables an analysis of their synchronization and impact on performance, with this synchronization measurable in terms of the mediating effects of capabilities. Previous studies (de Vasconcellos *et al*., 2019; 2021; Ameen *et al*., 2024) have examined such mediating effects. For example, de Vasconcellos *et al*. (2021) explain that organizational creativity acts as an input, capabilities function as processes, and performance serves as the outcome, highlighting that digital capabilities mediate the relationship between creativity and performance.

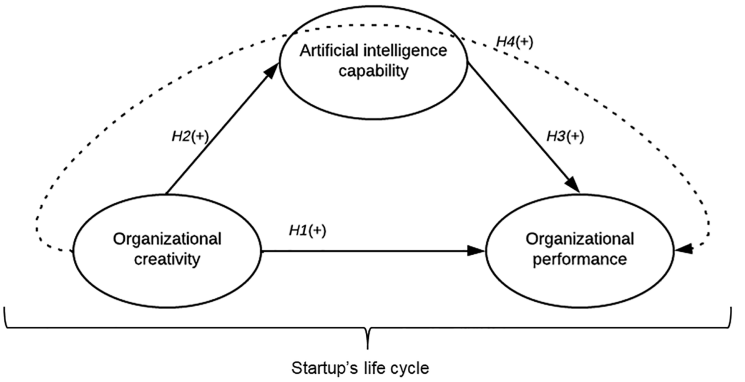
Similarly, Grilli and Pedota (2024) emphasize that integrating human creativity and AI in organizational management generates significant synergies, whereas AI alone, as an independent creator, has limited potential. Tran and Murphy (2023) further note that AI can indirectly support creative entrepreneurial activities, positively influencing performance. These interactions between creativity, AI, and performance can be explained by the mediating effect of AI Capability, which has not been fully explored, particularly in the TBS context. This analysis is essential for maintaining synchronization between resources and capabilities and ensuring that newly developed capabilities maximize performance (Sirmon *et al*., 2007, 2011). Therefore, we propose the following hypothesis:

H4. AI Capability mediates the relationship between organizational creativity and performance in TBS.

Figure 1 illustrates the research model and hypotheses, depicting the association between organizational creativity and organizational performance through AI Capability. The TBS life cycle, incorporated as a control variable, is tested in separate regressions for each phase.

In Figure 1, the life cycle illustrates the analysis of the initial, transition, and scalability phases (Picken, 2017a, b) of TBS and their impact on resource management processes, capability development, and performance outcomes.

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Figure 1. Conceptual model of the research

3. Methodological procedures

This study employs a quantitative approach, collecting data through a survey comprising 56 questions. Responses were measured on a seven-point Likert scale ranging from (1) strongly disagree to (7) strongly agree, with a midpoint option of (4) neither agree nor disagree. An additional eleven questions were included to gather information about company characteristics and respondents' roles. All questions were mandatory, mitigating the risk of bias arising from non-response errors or missing data (Cochran, 1977; Hair *et al.*, 2009).

Data analysis was conducted using multiple linear regression, with SPSS v.24 as the primary tool. To assess construct validity and reliability, AMOS v.16 was used to calculate factor loadings, Composite Reliability (CR), and Average Variance Extracted (AVE). Multiple regressions analysis is appropriate for this study as it addresses a single metric dependent variable presumed to relate to two or more metric independent variables (Hair *et al.*, 2009). This method also facilitates testing the mediation effects within a linear regression model, capturing the cause-and-effect sequence (Marôco, 2014).

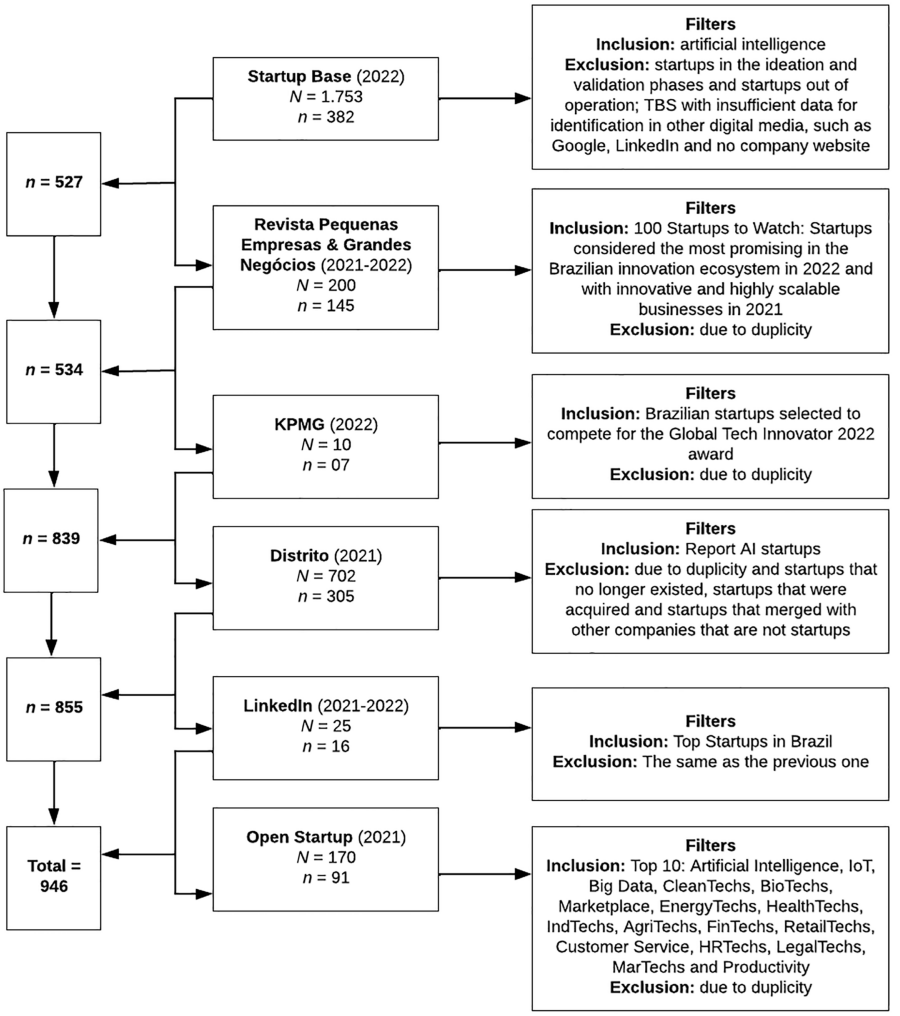
Additionally, the Sobel Test (see Sobel, 1982) was applied as a formal method to confirm the mediation effect (Baron and Kenny, 1986; Hall and Sammons, 2013; Marôco, 2014).

Data collection occurred between October 28, 2022, and January 16, 2023. The initial phase, from October 28 to November 10, 2022, involved a pilot test with 30 valid responses from managers and employees of technology-based startups (TBS), randomly selected from the same population as the main study. During this phase, CR and AVE values (see Hair *et al.*, 2009, 2021) were calculated, with CR > 0.80 and AVE >0.50, confirming the constructs' reliability and validity.

Based on these results, full data collection proceeded shortly after the pilot test, concluding on January 16, 2023. The initial sample of 30 respondents was retained in the final analysis, as no changes were made to the questionnaire following the pilot. Potential biases were addressed using both *ex ante* and *post hoc* approaches (see section 3.2).

To identify TBS operating in Brazil, we began with an existing database called Startup Base (ABStartups, 2022) and subsequently expanded it with additional data from sources such as Revista Pequenas Empresas & Grandes Negócios (REGN, 2021, 2022), KPMG (2022), Distrito (2021), LinkedIn (2021, 2022), and Open Startups (2022). Due to Brazil's General Personal Data Protection Law No. 13,709, of August 14, 2018, only non-sensitive data was accessed, ensuring compliance with privacy regulations while enabling the collection of relevant information. Figure 2 illustrates the process of consolidating the database, culminating in a total population of 946 TBS operating in Brazil and international markets.

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Figure 2. Stages of building the basis for data collection

Figure 2 also outlines the filters applied to implement the inclusion and exclusion criteria, with "*N*" denoting the total number of TBS and "*n*" representing the number selected after applying these filters.

The selection procedure followed the random sampling method (Cochran, 1977). With the database consolidated, the researcher responsible for data collection began following the identified TBS and sent connection invitations to managers and employees via LinkedIn. The questionnaires were distributed through LinkedIn and email to the entire population, accompanied by an invitation letter explaining the survey's objective—to investigate the performance of TBS.

In the invitation letter, we expressed gratitude for the respondents' voluntary participation and assured them that all information would remain consolidated and anonymized,

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guaranteeing the confidentiality. Data collection was supported by follow-up messages and phone calls, with three weekly reminders sent to improve the response rate.

To minimize the risk of common method bias, we sought multiple respondents from each TBS (Podsakoff *et al.*, 2003; Chang *et al.*, 2020). A total of 169 responses were obtained, of which three were excluded due to lack of variance, leaving 166 valid responses with no detected outliers (Cochran, 1977; Hair *et al.*, 2009).

To determine sample adequacy for multiple linear regression, we conducted the G*Power test (see Faul *et al.*, 2009). Using *a priori* F tests with an anticipated effect size of 0.30, 56 predictors, a desired statistical power level of 0.95 (1 - β err prob), and an alpha error probability of 0.05, the minimum required sample was 148. Our final sample of 166 exceeded this requirement, yielding a response rate of 17.55%.

Table 1 summarizes the main characteristics of the sample, including job titles, nationality, internationalization, revenue and life cycle phase.

The population of 946 TBS was concentrated in the Southeast and South regions, with São Paulo accounting for 53.2%, followed by Minas Gerais (9.4%), Paraná (7.5%), Santa Catarina (7.1%), and Rio Grande do Sul and Rio de Janeiro (5.9% each), representing 88.6% of the total. The sample of 166 respondents closely reflected this distribution, with São Paulo contributing 57.2%, followed by Santa Catarina (9.6%), Rio Grande do Sul (9.0%), Paraná (8.4%), Minas Gerais (6.0%), and Rio de Janeiro (2.4%), covering 92.8% of the sample. This similarity highlights the sample's representativeness and supports the generalizability of the findings.

Table 1. Sample characteristics

Characteristics	Categories	Number of respondents	Percentage	Cumulative
Job titles	C-Level	76	46%	46%
	Founders/Partners	16	10%	56%
	Directors	19	11%	67%
	Managers	30	18%	85%
	Analysts	03	2%	87%
	Data Scientists	06	4%	91%
	Business Developers	09	5%	96%
	Area Leaders	07	4%	100%
Nationality	Brazilian	160	96%	96%
	Other	06	4%	100%
Internationalization	Internationalized	39	23%	23%
	Operates only in Brazil	127	77%	100%
Revenue	Micro	35	21%	21%
	Small	81	49%	70%
	Medium	43	26%	96%
	Large	07	4%	100%
Phase	Initial	26	16%	16%
	Transition	60	36%	52%
	Scalability	80	48%	100%

Note(s): $N = 166$
Source(s): Authors' own work

3.1 Variable construction and measurement

The dependent variable, organizational performance, was measured using five items based on Lee and Choi (2003). The independent variable, organizational creativity, was also measured with five items derived from Lee and Choi (2003) and Scheibe and Gupta (2017). The mediating variable, AI Capability, was operationalized with 45 items (after validation) based on Mikalef and Gupta (2021) (see Appendix).

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To analyze the life cycle phases, respondents indicated their company's stage by selecting one of the following options: *Initial Phase*: the company focuses on defining and validating its business concept, market opportunity, offering, business model, and go-to-market strategy necessary to deliver the offering reliably to target customers at a profit. *Transition Phase*: the company begins to gain market traction, engages customers, and initiates product/service production and distribution, laying the foundation for a scalable business. *Scalability Phase*: the company grows rapidly, adding resources to scale profitably and establish market leadership (Picken, 2017a, b).

The life cycle phases were measured using dummy variables to control for stage-specific effects, as per Picken (2017a, b). For the initial phase, the dummy variable was defined as: initial (1), transition (0), scalability (0). Similarly, the transition phase was represented as: initial (0), transition (1), scalability (0), and the scalability phase as: initial (0), transition (0), scalability (1). This segmentation differentiates the resources that technology-based startups (TBS) possess and control, which influence AI Capability development and organizational performance (Carnes *et al.*, 2017; Sirmon *et al.*, 2011).

Beyond the life cycle phases of TBS, additional control variables were included to isolate *ceteris paribus* effects. These variables enhance the study's validity and reliability by accounting for factors that may influence resource management, AI Capability development, and organizational performance. The selected variables included:

Age and firm size: Age denotes the duration of a startup's existence since its foundation and can influence its ability to manage, utilize, hire, and negotiate resources (Zahra, 2021). *Firm size*, measured by the number of employees, affects resource availability (Carnes *et al.*, 2017), as well as creativity and innovation (Franco and Landini, 2022). Both factors ultimately influence capability building and organizational performance.

Revenue and Foreign Sales: *Revenue* measures the company's annual gross operating income or total revenue, while *Foreign Sales* measure the portion of revenue derived from international markets. Both variables indicate the company's financial capacity to invest in resources that enhance its capabilities and organizational performance.

Nationality: entrepreneurial culture varies by nationality (Zahra, 2021), influencing resource management practices and performance outcomes.

Internationalization: startups often internationalize early and rapidly, particularly digital startups (Cahen and Borini, 2020). Internationalization can impact exports, sales growth, customer base expansion, and other factors contributing to performance improvement.

3.2 Common method bias verification

To address common method bias, we used both *ex-ante* and *ex-post* approaches. *Ex-ante* measures included validating and translating the questionnaire through a back-translation process conducted by three PhD professors and two founders of award-winning technology-based startups (TBS) in Brazil. Additionally, we followed Podsakoff *et al.* (2003) by carefully designing the study procedures to minimize potential biases. *Ex-post* measures involved applying Harman's single-factor test, which indicated that 39.669% of the total variance was explained—well below the 50% threshold—suggesting that common method bias was not a concern (Chang *et al.*, 2020; Podsakoff *et al.*, 2003).

4. Results

4.1 Assessment of reliability and validity

Table 2 presents the indicator loadings, CR, Cronbach's Alpha, and AVE results.

The reliability of the indicators was confirmed. Although the questionnaire was fully applied after the pilot test, the indicator for the Inter-departmental Coordination construct, named "IC1," had a loading of 0.564. After analysis, it was excluded, resulting in an increase in the AVE of this construct from 0.50 to 0.54. All indicators had loadings above 0.60, except for

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Table 2. Validity and reliability of the constructs

Variable	Indicators	Loadings	Composite reliability	Cronbach's alpha	AVE
Organizational creativity	OC1	0.677	0.883	0.879	0.604
	OC2	0.694			
	OC3	0.832			
	OC4	0.765			
	OC5	0.895			
Artificial intelligence capability				0.832	0.660
Data	D1	0.690	0.843	0.836	0.518
	D2	0.644			
	D3	0.738			
	D4	0.770			
	D5	0.750			
Technology	T1	0.839	0.923	0.921	0.634
	T2	0.829			
	T3	0.671			
	T4	0.841			
	T5	0.727			
	T6	0.765			
	T7	0.881			
Basic resources	BR1	0.855	0.930	0.929	0.816
	BR2	0.932			
	BR3	0.921			
Inter-departmental coordination	IC2	0.757	0.876	0.871	0.543
	IC3	0.858			
	IC4	0.616			
	IC5	0.777			
	IC6	0.698			
	IC7	0.690			
Organizational change capacity	OCC1	0.761	0.861	0.853	0.513
	OCC2	0.551			
	OCC3	0.714			
	OCC4	0.806			
	OCC5	0.788			
	OCC6	0.643			
Risk proclivity	RP1	0.698	0.874	0.865	0.701
	RP2	0.930			
	RP3	0.867			
Technical skills	TS1	0.841	0.971	0.971	0.826
	TS2	0.941			
	TS3	0.981			
	TS4	0.947			
	TS5	0.864			
	TS6	0.861			
	TS7	0.917			
Business skills	BS1	0.918	0.960	0.960	0.753
	BS2	0.864			
	BS3	0.831			
	BS4	0.854			
	BS5	0.672			
	BS6	0.946			
	BS7	0.920			
	BS8	0.908			
Organizational performance	OP1	0.839	0.860	0.854	0.554
	OP2	0.795			
	OP3	0.757			
	OP4	0.733			
	OP5	0.571			

Source(s): Authors' own work

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OCC2 (0.551), which was retained to preserve the construct's content validity, following guidelines allowing retention of loadings above 0.50 (Hair *et al.*, 2021).

For CR, values between 0.80 and 0.90 are recommended (Hair *et al.*, 2021). However, empirical studies measuring AI Capability using the same construct (Li *et al.*, 2022; Mikalef and Gupta, 2021) reported similar results and considered values above 0.70 as indicative of good aggregation validity.

Cronbach's Alpha is a measure of internal consistency reliability that provides a conservative estimate of a construct's true reliability. Values between 0.80 and 0.90 are recommended (Hair *et al.*, 2021). The scores obtained were: organizational creativity ($\alpha = 0.879$), AI Capability ($\alpha = 0.832$), and organizational performance ($\alpha = 0.854$).

While the values for technical skills ($\alpha = 0.971$) and business skills ($\alpha = 0.960$) exceeded the recommended range (Hair *et al.*, 2021), their integration into the AI Capability construct ($\alpha = 0.832$) resulted in a score within the acceptable range. Therefore, the scales demonstrate high reliability and adequate internal consistency (>0.70) (Hair *et al.*, 2021; Marôco, 2014).

The AVE values for all constructs exceeded 0.50, indicating that each construct explain 50% or more of the variance in its respective indicators (Hair *et al.*, 2021).

Discriminant validity assesses whether a construct is empirically distinct from other (Hair *et al.*, 2021). We evaluated it using the Fornell-Larcker criterion (see Table 3). Discriminant validity is confirmed when the square roots of the AVE values are greater than the Pearson correlations between the constructs (Fornell and Larcker, 1981).

The results in Table 3 confirm the discriminant validity of the constructs, as the square roots of the AVE values exceed the Pearson correlations shown in Table 4.

With all measurement model requirements confirmed, we proceeded to test the hypotheses.

Table 3. Fornell-Larcker criterion

Variables	AVE	1	2	3
1 Organizational creativity	0.604	*0.777*		
2 AI capability	0.660	0.446	*0.813*	
3 Organizational performance	0.554	0.224	0.327	*0.745*

Source(s): Authors' own work

Table 4. Descriptive statistics with correlation coefficients

Variables	Mean	Standard deviation	1	2	3
1 Organizational creativity	6.124	0.818	1		
2 Artificial intelligence capability	5.094	1.160	*0.446*	1	
3 Organizational performance	4.980	1.174	0.224	*0.327*	1

Note(s): Pearson Correlation (two-tailed). $N = 166$
Source(s): Authors' own work

4.2 Hypothesis testing

Before testing the hypotheses, we examined the descriptive statistics, including means, standard deviations, and correlations among the constructs (see Table 4).

The results indicate that organizational creativity is positively correlated with organizational performance ($r = 0.224$, $p = 0.004$) and AI Capability ($r = 0.446$, $p < 0.000$). Additionally, AI Capability is positively correlated with organizational performance ($r = 0.327$, $p < 0.000$). These results shows that all constructs vary in the

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same direction, with the intensity of the associations being moderate ($r > 0$ e $0{,}25 \leq |r| < 0{,}5$) (Marôco, 2014).

The mediation test follows three steps (Baron and Kenny, 1986; Marôco, 2014; Demming *et al.*, 2017; van den Berg, 2023):

Step 1. Analyze path *c* to examine the total effect of organizational creativity on performance (H1+) and subsequently verify if *c* is significant and different from *c'*.

Step 2. Analyze path *a* to examine the effect of organizational creativity on AI Capability (H2+) and check if *a* is significant.

Step 3. Analyze path *b* to examine the effect of AI Capability on performance (H3+) and path *c'* to examine the direct effect of organizational creativity on performance.

These steps help verify whether *b* and *c'* are significant and if *c'* is closer to zero than *c* (total effect), indicating a mediation effect (H4+).

We performed these steps to analyze the effects (*a*, *b*, *c*, and *c'*) (Baron and Kenny, 1986; Demming *et al.*, 2017) across the three life cycle phases of technology-based startups (TBS): initial, transition, and scalability. Control variables were included in all regression models, and the results are presented below.

Table 5 shows the total effect of organizational creativity on organizational performance in TBS.

Hypothesis 1 is supported, showing that organizational creativity is positively associated with performance across all TBS life cycle phases. However, the initial and transition phases show negative effects, while the scalability phase demonstrates a positive effect.

To test H1, we performed regression analyses for each life cycle phase: initial (Model 1), transition (Model 2), and scalability (Model 3). The regression results were as follows: Model 1 ($\beta = 0.205$, $p = 0.009$); Model 2 ($\beta = 0.202$, $p = 0.012$), and Model 3 ($\beta = 0.219$, $p = 0.003$), indicating that organizational creativity drives performance in TBS, especially as they advance to more mature phases in their life cycle.

However, in Models 1 and 2, the coefficients for the initial and transition phases were negative for performance (Model 1: $\beta = -0.342$, $p = 0.000$; Model 2: $\beta = -0.204$, $p = 0.009$). This suggests that, although creativity contributes positively, specific challenges in these phases—such as defining and validating the business concept during the initial phase and

Table 5. Results of the *c* effect (total effect)

	Model 1. initial		Model 2. transition		Model 3. scalability	
	Standardized coefficients/beta and variance inflation factor					
	B	VIF	β	VIF	β	VIF
Age	0.037	1.124	0.119	1.031	0.002	1.124
Firm size	−0.070	1.510	−0.128	1.495	−0.101	1.492
Revenue	−0.004	1.102	0.072	1.089	0.049	1.075
External sales	−0.061	1.470	−0.002	1.451	−0.034	1.450
Nationality	−0.100	1.720	−0.104	1.720	−0.085	1.723
Internationalization	−0.023	1.221	−0.026	1.223	−0.046	1.226
Phase	−0.342***	1.160	−0.204**	1.047	0.450***	1.121
Organizational creativity	0.205**	1.136	0.202*	1.137	0.219**	1.139

Note(s): Dependent variable: **Organizational Performance**
Model 1: $R^2 = 0.180$; F = 4.331; Sig. = 0.000; Non-Standardized Coefficients (N-SC) ($\beta_c = 0.294$)
Model 2: $R^2 = 0.119$; F = 2.660; Sig. = 0.009; N-SC ($\beta_c = 0.290$)
Model 3: $R^2 = 0.260$; F = 6.913; Sig. = 0.000; N-SC ($\beta_c = 0.315$)
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$
Source(s): Authors' own work

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building the company's foundation for a scalable enterprise during the transition phase (Picken, 2017a, b)—involving resource constraints (Carnes *et al.*, 2017) and time limitations (Zahra, 2021), may initially have negative effects on performance (Picken, 2017b; Carnes *et al.*, 2017; Sirmon *et al.*, 2011).

In contrast, the scalability phase showed a positive effect on performance (Model 3: $\beta = 0.450$, $p = 0.000$), indicating that as firms grow, creativity becomes even more impactful, driving higher performance. These findings highlight the need to foster organizational creativity across all life cycle stages.

Table 6 presents the effects of organizational creativity on AI Capability in TBS.

Hypothesis 2 is supported, showing that organizational creativity is positively associated with AI Capability across all TBS life cycle phases. However, firm size negatively influences the development of AI Capability.

To test H2, we conducted regression analyses for each life cycle phase: initial (Model 4), transition (Model 5), and scalability (Model 6). The regression results were as follows: Model 4 ($\beta = 0.408$, $p = 0.000$), Model 5 ($\beta = 0.405$, $p = 0.000$) and Model 6 ($\beta = 0.410$, $p = 0.000$). These findings underscores the critical role of organizational creativity as a foundation for adopting and advancing technologies such as AI, highlighting its consistent and significant role as a key antecedent of AI Capability in TBS across different life cycle phases.

Interestingly, the firm size showed a negative association with AI Capability across all phases (Model 4: $\beta = -0.170$, $p = 0.049$; Model 5: $\beta = -0.187$, $p = 0.030$; Model 6: $\beta = -0.183$, $p = 0.033$). This suggests that larger firms may encounter challenges in fostering high levels of strategic and workforce agility (Ameen *et al.*, 2024; Franco and Landini, 2022), both of which are linked to stronger employee commitment and motivation. Such limitations can hinder creativity and innovation (Acar *et al.*, 2019), which are critical for building AI Capability (Ameen *et al.*, 2024). Conversely, smaller TBS demonstrate greater agility and benefit from streamlined processes and flexibility, allowing them to leverage creativity more effectively for capability development (Sirmon *et al.*, 2011; Carnes *et al.*, 2017).

Finally, Table 7 presents the effects of AI Capability on performance and the direct effect of organizational creativity on performance. This allows us to analyze the mediation role of AI Capability in linking creativity to performance in TBS.

Hypothesis 3 is supported, showing that AI Capability is positively associated with performance across all TBS life cycle phases. However, while the initial and transition phases show negative effects on performance, the scalability phase demonstrates a positive effect. The

Table 6. Results of the *a* effect

	Model 4. initial		Model 5. transition		Model 6. scalability	
	Standardized coefficients/beta and variance inflation factor					
	β	VIF	β	VIF	β	VIF
Age	−0.019	1.124	0.016	1.031	−0.014	1.124
Firm size	−0.170*	1.510	−0.187*	1.495	−0.183*	1.492
Revenue	−0.026	1.102	−0.005	1.089	−0.007	1.075
External sales	−0.071	1.470	−0.053	1.451	−0.059	1.450
Nationality	−0.109	1.720	−0.113	1.720	−0.107	1.723
Internationalization	0.084	1.221	0.086	1.223	0.080	1.226
Phase	−0.126	1.160	−0.018	1.047	0.108	1.121
Organizational creativity	0.408***	1.136	0.405***	1.137	0.410***	1.139

Note(s): Dependent variable: **AI Capability**
Model 4. $R^2 = 0.241$; F = 6.228; Sig. = 0.000; N-SC ($\beta_a = 0.579$); Standard Error ($SE_a = 0.105$)
Model 5: $R^2 = 0.227$; F = 5.778; Sig. = 0.000; N-SC ($\beta_a = 0.575$); $SE_a = 0.106$
Model 6: $R^2 = 0.237$; F = 6.111; Sig. = 0.000; N-SC ($\beta_a = 0.582$); $SE_a = 0.106$
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$
Source(s): Authors' own work

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Table 7. Results of the *b* and *c*' (direct effect) effects

	Model 7. initial		Model 8. transition		Model 9. scalability	
	Standardized coefficients/beta and variance inflation factor					
	B	VIF	β	VIF	β	VIF
Age	0.041	1.124	0.115	1.031	0.005	1.125
Firm size	−0.031	1.548	−0.078	1.541	−0.061	1.536
Revenue	0.002	1.103	0.073	1.089	0.050	1.075
External sales	−0.045	1.477	0.012	1.455	−0.022	1.455
Nationality	−0.075	1.736	−0.074	1.737	−0.061	1.738
Internationalization	−0.042	1.230	−0.049	1.233	−0.064	1.234
Phase	−0.313***	1.181	−0.199**	1.048	0.427***	1.137
Organizational creativity	0.111	1.356	0.093	1.349	0.129	1.359
AI capability	0.229**	1.317	0.269**	1.294	0.220**	1.311

Note(s): Dependent variable: **Organizational Performance**
Model 7. $R^2 = 0.220$; F = 4.888; Sig. = 0.000; N-SC ($\beta_{c'} = 0.160$; $\beta_b = 0.232$); $SE_b = 0.082$
Model 8: $R^2 = 0.175$; F = 3.681; Sig. = 0.000; N-SC ($\beta_{c'} = 0.134$; $\beta_b = 0.272$); $SE_b = 0.084$
Model 9: $R^2 = 0.297$; F = 7.339; Sig. = 0.000; N-SC ($\beta_{c'} = 0.185$; $\beta_b = 0.223$); $SE_b = 0.078$
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$
Source(s): Authors' own work

non-significance of organizational creativity confirms the full mediation effect of AI Capability across all phases of TBS, supporting H4. These results highlight the increasing importance of AI Capability as startups progress through their life cycle.

To test H3, we conducted regression analyses for each life cycle phase: initial (Model 7), transition (Model 8), and scalability (Model 9). The regression results were as follows: Model 7 ($\beta = 0.229$, $p = 0.005$), Model 8 ($\beta = 0.269$, $p = 0.001$), and Model 9 ($\beta = 0.220$, $p = 0.005$). These findings demonstrate that AI Capability plays a crucial role in enhancing organizational performance across all phases of the TBS life cycle.

However, in Models 7 and 8, the coefficients for the initial and transition phases were negative for performance (Model 7: $\beta = -0.313$, $p = 0.000$; Model 8: $\beta = -0.199$, $p = 0.008$). These results suggest that, despite the positive role of AI Capability, the same specific challenges identified in Models 1 and 2 may initially have negative effects on performance.

Although previous studies link creativity, AI Capability, and performance positively, our research takes a novel approach by analyzing TBS across different development phases. Negative coefficients in the initial and transition stages are expected during these phases (Picken, 2017b). A notable example illustrating these negative associations is Uber, which operated at a loss from its founding in 2009 until reporting its first profit in 2023 (Financial Times, 2024).

In contrast, Model 9, representing the scalability phase, shows a positive and significant coefficient ($\beta = 0.427$, $p = 0.000$). This indicates that as startups mature, AI Capability delivers greater performance benefits, underscoring its critical role throughout the life cycle, particularly during scalability.

To test Hypothesis 4 and analyze the mediation effect, we followed van den Berg's (2023) recommendation of using unstandardized coefficients (N-SC) and *p*-values paths *a*, *b*, *c* and *c'*. The N-SC values for *c'* ($\beta = 0.160$; $\beta = 0.134$, and $\beta = 0.185$) are closer to zero than those N-SC values for *c* ($\beta = 0.294$; $\beta = 0.290$, and $\beta = 0.315$), while paths *a*, *b* and *c* are significant ($p < 0.05$) and path *c'* is not significant ($p > 0.05$). These results confirm that AI Capability fully mediates the relationship between organizational creativity and performance, supporting H4 (Baron and Kenny, 1986; Demming *et al.*, 2017; van den Berg, 2023).

We also conducted the Sobel Test (Sobel, 1982; Baron and Kenny, 1986; Hall and Sammons, 2013) using the N-SC values and standard errors of paths *a* and *b*. For the initial phase ($\beta a = 0.579$; $\beta b = 0.232$; SEa = 0.105; SEb = 0.082), the Sobel Test statistic was 2.517, with one-tailed $p = 0.005$ and two-tailed $p = 0.011$. For the transition phase ($\beta a = 0.575$;

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βb = 0.272; SEa = 0.106; SEb = 0.084), the Sobel Test statistic was 2.780, with one-tailed $p = 0.002$ and two-tailed $p = 0.005$. For the scalability phase (βa = 0.582; βb = 0.223; SEa = 0.106; SEb = 0.078), the Sobel Test statistic was 2.535, with one-tailed $p = 0.005$ and two-tailed $p = 0.011$. These results confirm the full mediation effect of AI Capability, supporting H4.

Finally, collinearity was assessed using the Variance Inflation Factor (VIF), and all values are below 2, indicating no multicollinearity concerns (Marôco, 2014; see Tables 5–7).

5. Discussion

This study highlights the interrelation between organizational creativity, AI Capability, and performance in technology-based startups (TBS) across life cycle phases, illustrating the process by which TBS develop AI Capability to enhance performance. Supported by four hypotheses, the findings advance management theory by deepening the understanding of AI Capability development and its strategic implications. Grounded in Resource Orchestration Theory (ROT), we address key managerial challenges related to AI resource management and capability building.

First, the results confirm that organizational creativity positively influences TBS performance, supporting H1 and aligning with prior research (de Vasconcellos *et al.*, 2021; 2024; Lee and Choi, 2003; Mikalef and Gupta, 2021; Weinzimmer *et al.*, 2011). Creativity is especially impactful during the scalability phase, where it drives measurable performance gains.

Second, organizational creativity significantly contributes to AI Capability in TBS, supporting H2. This finding aligns with literature positioning organizational creativity as a foundational element for capability development (Javidan, 1998; de Vasconcellos *et al.*, 2019; 2021; 2024) and reinforces the ROT framework (Sirmon *et al.*, 2007, 2011). Creativity plays a strategic role in integrating AI resources—tangible, intangible, and human—into AI Capability. For larger TBS, workforce agility can enhance this relationship, making resource orchestration more effective (Ameen *et al.*, 2024; Franco and Landini, 2022; Acar *et al.*, 2019).

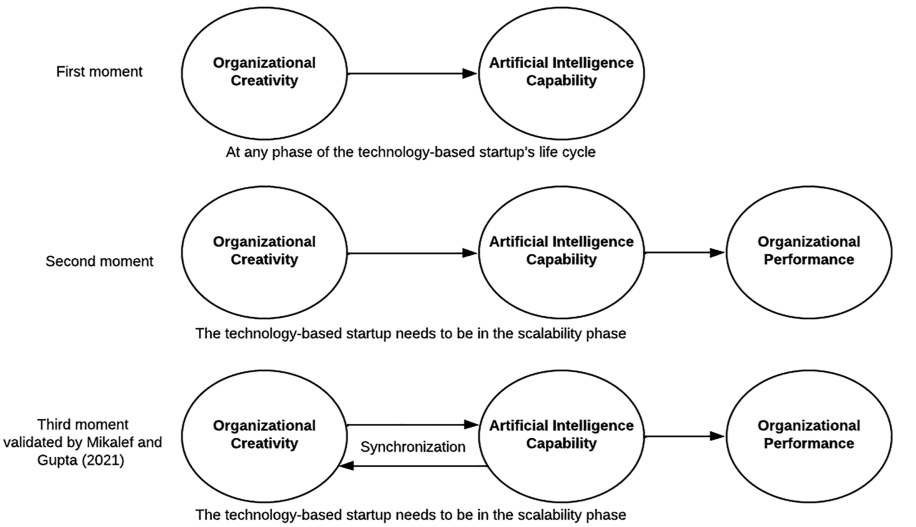
Third, AI Capability is positively associated with TBS performance, supporting H3. This finding aligns with previous studies (Chowdhury *et al.*, 2023; Mikalef and Gupta, 2021; Wamba-Taguimdje *et al.*, 2020) and emphasizes AI Capability's role in driving competitive advantage (Andersén, 2021; Sund *et al.*, 2018). Its impact is most pronounced during the scalability phase, where TBS excel in bundling resources to enhance performance (Sirmon *et al.*, 2007, 2011).

Fourth, the findings demonstrate that AI Capability fully mediates the relationship between creativity and performance in TBS, supporting H4. This mediation underscores a cause-and-effect sequence in which creativity drives AI Capability, which then enhances performance (Marôco, 2014). This result aligns with studies highlighting capabilities as mediators between resources and performance (de Vasconcellos *et al.*, 2021) and emphasizes creativity's foundational role as a "resource-block" for capability development (Javidan, 1998; de Vasconcellos *et al.*, 2019; 2024).

5.1 Theoretical implications

Our findings advance the understanding of AI Capability development, addressing Mikalef and Gupta's (2021) call for a process-based perspective. By examining technology-based startups (TBS) across life cycle phases and stages of AI implementation, we identify a distinct pathway—"organizational creativity → AI Capability → performance"—where creativity catalyzes the formation of AI Capability, which, in turn, drives performance. This trajectory underscores the effectiveness of AI resource orchestration in TBS contexts and aligns with Resource Orchestration Theory (ROT), which posits that capabilities intervene in the relationship between resources and performance, fostering continuous improvement through the synchronization of resources and capabilities (Sirmon *et al.*, 2007, 2011). Our study complements prior research (Mikalef and Gupta, 2021; Li *et al.*, 2022) by elucidating three key moments in this process (Figure 3).

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Figure 3. AI capability implementation trajectory based on organizational creativity

Figure 3 illustrates the three moments where TBS strategically leverage organizational creativity to implement AI solutions, develop AI Capability, and achieve sustained performance.

5.1.1 First moment: creativity drives the development of AI capability. Creativity serves as a foundational resource for acquiring, developing, and structuring AI-related resources (Glaveanu and Taillard, 2018; Zahra, 2021), transforming them into AI Capability through bundling processes like pioneering (Sirmon *et al.*, 2007, 2011). This moment highlights the "organizational creativity → AI Capability" pathway (H2 supported), emphasizing creativity's strategic importance in initiating capability development across TBS life cycle phases.

These findings underscore the critical role of organizational creativity in resource orchestration, particularly in structuring and bundling processes. Investing in creative practices and workforce agility enables TBS to enhance AI Capability, sustain innovation, and maintain competitive advantages in dynamic markets.

5.1.2 Second moment: AI capability mediates the creativity-performance relationship. AI Capability enhances the impact of creativity on performance by facilitating the integration of AI technologies into operations (Abaddi, 2023, 2024), with synchronization between creativity (structuring) and AI Capability (bundling) amplifying outcomes, particularly during scalability phases. The mediation effect of AI Capability (H4 supported) aligns with the perceived usefulness of AI technologies, promoting their adoption and seamless integration into business processes (Abaddi, 2024).

While TBS in initial and transition phases can orchestrate creativity and AI Capability, these phases are less favorable for performance due to resource constraints. However, during scalability, synchronization becomes more effective, driving higher performance and reinforcing the "organizational creativity → AI Capability → performance" pathway.

This moment also contributes to ROT by proposing synchronization as a vital process in resource management. By aligning resources and capabilities and integrating their interactions, synchronization enhances organizational outcomes and supports continuous adaptation and operational transformation.

5.1.3 Third moment: creativity and AI capability co-evolve to sustain performance. AI Capability evolves to influence and enhance creativity, creating a feedback loop that drives

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innovation and adaptability (Mikalef and Gupta, 2021; Li *et al.*, 2022). In this third moment, AI Capability not only mediates the impact of creativity on TBS performance but also begins to influence and enhance creativity itself. This reciprocal relationship gives rise to the "AI Capability → organizational creativity → performance" pathway (Mikalef and Gupta, 2021), illustrating the dynamic interplay between resources and capabilities.

This co-evolution fosters cyclical synchronization, enabling TBS to refine strategies and align resources with market demands (Sirmon *et al.*, 2007, 2011). The reciprocal relationship between creativity and AI Capability strengthens performance, positioning TBS for sustained growth and competitive advantage (Grilli and Pedota, 2024; Zahra, 2021).

At this stage, AI technologies (Abaddi, 2023), supported by tangible, intangible, and human AI resources (Mikalef and Gupta, 2021; Li *et al.*, 2022), exhibit artificial autonomy (Abaddi, 2024). However, for AI autonomy to generate meaningful creative outputs, organizational creativity must guide the development of AI resources (Grilli and Pedota, 2024). This ensures that AI Capability remains rooted in creativity, fostering a virtuous cycle where AI Capability and creativity co-evolve to sustain innovation and performance.

These findings highlight the co-evolutionary dynamics of creativity and AI Capability, contributing to ROT by introducing synchronization as a crucial process. This refined perspective enables a deeper understanding of how capabilities evolve and interact with resources to achieve strategic objectives, offering a robust framework for sustained innovation and competitive advantage in TBS contexts.

5.2 Practical, public policy, and social implications

This discussion highlights practical implications for managers, policymakers, and society. These insights should be tailored to each technology-based startup's (TBS) unique characteristics, resources, and strategic objectives.

First, TBS managers should maintain lean, highly qualified teams. As the TBS grows, it is crucial to promote workforce agility and minimize bureaucracy, which can stifle creativity and hinder the development of AI Capability. For example, as the number of employees increases, managers can organize smaller, autonomous teams that operate with greater creativity and agility. This approach allows multiple creative teams to lead specific AI-related initiatives, which, when integrated, can accelerate and enhance the implementation of AI Capability.

Second, managers should address the specific challenges of the initial and transition phases by adjusting strategies and allocating resources accordingly. For instance, during the early phases, they should focus on fostering creativity and developing AI resources while managing limited resources and laying a solid foundation for future scalability. Another example is to provide AI and machine learning training for employees and invest in AI tools that integrate seamlessly into business processes, such as data analytics, process automation, and custom AI solutions. These strategies can help mitigate resource constraints in the early stages and enhance organizational performance.

Third, managers should foster creativity within TBS by cultivating an environment that encourages innovation and idea generation. For instance, they can implement internal programs such as hackathons, creative workshops, and brainstorming sessions to stimulate employee creativity. Additionally, organizing AI-driven innovation workshops with cross-functional teams can help develop creative solutions and explore the intersection of creativity and AI. Another effective practice is leveraging AI-powered market analysis tools to identify emerging trends. These creativity-focused strategies not only generate innovative ideas but also strengthen AI Capability, ultimately enhancing organizational performance.

Fourth, managers should continuously assess the impact of creativity and AI Capability on performance and adapt strategies as necessary. For instance, they can establish feedback loops to monitor and enhance both creativity and AI Capability. These iterative cycles enable TBS to quickly respond to market and technological changes. Additionally, managers can encourage

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teams to learn from each cycle and share insights, further reinforcing AI Capability and organizational creativity.

For policymakers, our findings suggest three key implications. First, they should implement policies that foster innovation and the development of AI technologies in TBS. Examples include providing subsidies, research and development grants, tax incentives, and establishing public-private partnerships to fund research hubs dedicated to AI and innovation. Such initiatives can help TBS integrate AI and creativity, enhancing their competitiveness and capacity for innovation. Second, creating policies that support the growth and sustainability of TBS is essential for maximizing their innovative potential. This could involve establishing startup incubators and accelerators that offer technical, financial, and mentoring support to help these companies grow and thrive. Third, investing in AI education and training is crucial for preparing the workforce to meet future technological demands. Policymakers could introduce educational programs focused on entrepreneurship and AI technologies in technical schools and universities, as well as provide training courses for professionals already in the workforce.

Our findings also have broader societal implications. First, by demonstrating that AI extends human creativity, public attitudes toward AI acceptance could improve. Second, enhancing AI education and training can elevate digital literacy within startups, contributing to a stronger economy by creating new jobs and boosting productivity. Third, by showing how creativity and AI can thrive even in resource-limited environments, this study may inspire other countries to invest more in technology development, promoting greater digital and technological inclusion, particularly in emerging markets and developing countries.

5.3 Limitations and future research directions

This study has several limitations that suggest avenues for future research. As a cross-sectional study, it could not establish a temporal sequence between the life cycle phases of technology-based startups (TBS). While we isolate each phase in the regression analyses to better understand the orchestration of creativity and the development of AI Capability, it remains challenging to argue that these phases are strictly sequential. Longitudinal studies could address this limitation by demonstrating causality and tracking changes over time, offering a more robust understanding of the underlying dynamics.

This study validated the AI Capability scale for the context of TBSs, with a randomly drawn sample from an emerging country. While this provides a strong foundation for understanding AI Capability, the unique characteristics of this context may limit the generalizability of the findings. Future research could apply this scale in diverse geographic regions and industrial sectors to further analyze AI Capability development and broaden its applicability. Comparative studies across different countries and industries could also enhance the theoretical and practical relevance of the results.

This study relied exclusively on a quantitative approach, which, while offering statistically significant insights, does not capture the contextual depth that qualitative methods could provide. Future research could adopt qualitative approaches, such as interviews with managers, to explore how TBS navigate challenges across different life cycle phases and how firm size influences the development of AI Capability. Building on our findings, qualitative studies could also investigate how organizational creativity is used to acquire and develop tangible, intangible, and human AI resources, as well as how these resources are integrated into AI Capability. Additionally, examining the third moment, where creativity and AI Capability synchronize, could provide deeper insights into their continuous flow of mutual influence and improvement within TBS.

Finally, future research could investigate cultural influences on the relationship between organizational creativity and AI Capability across different markets.

6. Conclusion

This study demonstrated how technology-based startups (TBS) leverage organizational creativity to build AI Capability and enhance performance across different stages of their life

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cycle. Using multiple linear regressions and the Sobel test, we developed and tested four research hypotheses, all of which were supported. Our findings reveal the pathway TBS follow through structuring and bundling resources to develop AI Capability grounded in organizational creativity. Importantly, the synchrony analyzed through the mediating effect of AI Capability transforms this pathway into a recursive cycle, initiating a new management process where resources and capabilities co-evolve at the intersection of creativity and AI.

The results confirm that organizational creativity is positively associated with AI Capability, and both are linked to performance through the full mediation effect of AI Capability. At various life cycle stages, TBS can harness the creativity embedded in their resource portfolios to develop AI Capability, positioning themselves for performance gains, particularly during the scalability phase.

Building on these findings, we propose a sequence of moments that provides a procedural and dynamic perspective on the integration of organizational creativity and AI Capability to enhance performance in TBS. This perspective aligns with Resource Orchestration Theory, complementing and validating prior research, while emphasizing that resource management in TBS is an iterative and evolutionary process.

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Appendix

Table A1. Measurement of variables

Construct	Item
Organizational creativity (OC, five items) Lee and Choi (2003), Scheibe and Gupta (2017), Mikalef and Gupta (2021)	(OC1): Our organization has produced many novel and useful ideas (services/products) (OC2): Our organization fosters an environment conducive to our own ability to produce novel and useful ideas (services/products) (OC3): Our organization spends much time producing novel and useful ideas (services/products) (OC4): Our organization considers producing novel and useful ideas (services/products) as important activities (OC5): Our organization actively produces novel and useful ideas (services/products)
Artificial intelligence capability (AI capability, 45 items) (adapted from Mikalef and Gupta, 2021)	Construct is formed by integrating: tangible resources, intangible resources and human resources
Tangible resources	
Data (D, five items)	(D1): Our organization has access to a large amount of data analysis (D2): Our organization integrates data from multiple internal sources, storing it in an accessible data repository controlled by the organization itself (D3): Our organization integrates external data with internal data to facilitate strategic-level value analysis of our business environment (D4): Our organization is capable of preparing and cleaning data with the aim of using Artificial Intelligence (AI) models efficiently and evaluating data for accuracy (D5): Our organization is able to obtain data at the right level of granularity to produce meaningful insights
Technology (T, seven items)	(T1): Our organization explores and adopts cloud-based services to process data and perform AI and machine learning (T2): Our organization has the necessary processing power to support AI applications (e.g. CPUs, GPUs) (T3): Our organization invests in network infrastructure (e.g. enterprise networks) that supports the efficiency and scale of applications (scalability, high bandwidth, and low latency) (T4): Our organization explores and adopts parallel computing approaches for AI data processing (T5): Our organization invests in advanced cloud services to allow complex AI abilities on simple API calls (e.g. Microsoft Cognitive Services, Google Cloud Vision) (T6): Our organization invests in scalable data storage infrastructures (T7): Our organization explores AI infrastructure to ensure that data is secured from end to end with state-of-the-art technology
Basic resources (BR, three items)	(BR1): In our organization, the AI initiatives are adequately funded (BR2): In our organization, the AI project has enough team members to get the work done (BR3): In our organization the AI project is given enough time for completion

(*continued*)

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Table A1. Continued

Construct	Item
Intangible resources	
Inter-departmental coordination (IC, six items)	*(IC1): In our organization, departments (e.g. marketing, R&D, manufacturing, information technology, and sales) are engaged in internal collaborative activities* (*deleted item*) (IC2): In our organization, departments have collective goals (IC3): In our organization, departments work as teamwork (IC4): In our organization, departments have the same view on the adoption of digital technologies (IC5): In our organization, departments have mutual understanding (IC6): In our organization, departments share information (IC7): In our organization, departments share resources
Organizational change capacity (OCC, six items)	(OCC1): Our organization is able to anticipate and plan for the organizational resistance to change (OCC2): Our organization considers business process re-engineering to revamp key processes by incorporating AI advancements (OCC3): Our organization recognizes the need to manage change (OCC4): Our organization is capable to communicate the reasons for change to its members (OCC5): Our organization is able to make the necessary changes in human resources policies for process re-engineering (OCC6): In our organization, senior management commits to new values as they incorporate advances in AI
Risk proclivity (RP, three items)	(RP1): In our organization, we have a strong proclivity for high-risk projects (with chances of very high returns) (RP2): In our organization we take bold and wide-ranging acts to achieve firm objectives (RP3): In our organization, we typically adopt a bold, aggressive posture in order to maximize the probability of exploiting potential opportunities
Human resources	
Technical skills (TS, seven items)	(TS1): Our organization has employees with the right technical skills to support AI work (TS2): In our organization, data scientists are very capable of using AI technologies (e.g. machine learning, natural language processing, deep learning) (TS3): In our organization, data scientists have the right skills to accomplish their jobs successfully (TS4): In our organization, data scientists are effective in data analysis, processing, and security (TS5): In our organization, data scientists are provided with the required training to deal with AI applications (TS6): Our organization hires data scientists who have the AI skills we are looking for (TS7): In our organization, data scientists have suitable work experience to fulfill their jobs

(*continued*)

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Table A1. Continued

Construct	Item
Business skills (BS, eight items)	(BS1): In our organization, managers are able to understand business problems and direct AI initiatives to solve them (BS2): In our organization, managers are able to work with data scientists, other employees, and customers to determine the opportunities that AI might bring to our organization (BS3): In our organization, managers have a good sense of where to apply AI (BS4): In our organization, the AI manager has strong leadership skills (BS5): In our organization, managers are able to anticipate the future business needs of functional managers, suppliers, and customers (BS6): In our organization, managers are able to proactively design AI solutions to support the needs of functional managers, suppliers, and customers (BS7): In our organization, managers are capable of coordinating AI-related activities in ways that support the organization, suppliers, and customers (BS8): Our organization has strong leadership to support AI initiatives, and managers demonstrate ownership and commitment to AI projects
Organizational performance (OP, five items) Lee and Choi (2003), Mikalef and Gupta (2021)	(OP1): Compared to our key competitors, our organization is more successful (OP2): Compared to our key competitors, our organization has a greater market share (OP3): Compared to our key competitors, our organization is growing faster (OP4): Compared to our key competitors, our organization is more profitable (OP5): Compared to our key competitors, our organization is more innovative

Source(s): Adapted from Lee and Choi (2003), Scheibe and Gupta (2017), and Mikalef and Gupta (2021)

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